

STATSIG lecture 7

Sigma-point Kalman filter

for non-linear systems

(aka. UKF, Unscented Kalman Filter)

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

Sigma Point Kalman Filter (aka. UKF, Unscented Kalman Filter)

<u>Non-linear system</u>	<u>Process</u>	$\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{w}_k,$	$\mathbf{w}_k \sim (0, \mathbf{Q}_k)$
	<u>Measurement</u>	$\mathbf{y}_k = \mathbf{h}(\mathbf{x}_k) + \mathbf{v}_k,$	$\mathbf{v}_k \sim (0, \mathbf{R}_k)$

- EKF** linearizes non-linearities around currently estimated states
only correct up to first order of Taylor series for non-linearities
requires calculation of Jacobians
only works for weak non-linearities or EKF might diverge (giving meaningless estimates)
- UKF** tries to improve state estimation for non-linear systems
probabilistic approach somewhat similar to Monte Carlo methods
no approximation to non-linearities, are used directly in algorithm

First: More general problem (Unscented Transform, UT)

Propagation of moments through non-linearities using sigma (test) points

General problem

Given random vector $X \in \mathbb{R}^n$ with mean $\bar{x}$ and covariance $P_x = E[(X - \bar{x})(X - \bar{x})^T]$

Non-linear vector function of vector variable $y = f(x)$

Find for new random vector $Y = f(X) \in \mathbb{R}^n$ the mean $\bar{y}$ and covariance $P_y = E[(Y - \bar{y})(Y - \bar{y})^T]$

EKF approach $\bar{y} = f(\bar{x})$

Due to Jensens inequality this makes systematic errors

By making a multivariate Taylor-series development of the non-linear function it can be seen that the **EKF approach only estimates state and covariance matrix** (both used in the EKF algorithm) **correctly up to first order** (not so strange since it uses linearization)

UT, Unscented Transform approach

Basic idea: Define $2n$ x-sigma-points, $x^{(i)}$, around the mean $\bar{x}$
Propagate the x-sigma-points perfectly through the non-linear function (no linearization or approximation)
This gives $2n$ y-sigma-points, $y^{(i)}$
Potentially define weights, $w^{(i)}$, (often not applied)
Estimate mean $\bar{y}$ and covariance matrix P_y from y-sigma-points

Details on the blackboard ...

The x-sigma-points are chosen so that $E[x^{(i)}] = \bar{x}$, $Cov[x^{(i)}] = P_x$

This approach estimates state and covariance matrix correctly up to third order
(EKF only up to first order)

Finally: UT approach for state estimation used in Kalman filter algorithm
See pdf-file UKF algorithm